

Power BI Exploratory Data Analysis

SMA, Visualization and Interactive Dashboard

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Programme: B.Sc. Data Science and AI

Tool Used: Microsoft Power BI Desktop

Dataset: Shopify Stock Dataset

1. Introduction:

Week 4 focused on performing Exploratory Data Analysis (EDA) on the Shopify stock dataset using Microsoft Power BI. The work built on the structured data model and DAX measures created in the previous week. The analysis included calculating 20-day and 50-day Simple Moving Averages (SMA), creating a moving average crossover signal, developing interactive price and trading-volume visualizations, configuring date and parameter filters, and creating a custom stock tooltip.

The main objective was to transform the Shopify stock data into an interactive analytical dashboard that supports the examination of price trends, moving averages, trading volume and selected stock metrics over time.

2. Calculating 20-Day Simple Moving Average:

A 20-day Simple Moving Average (SMA) was created using DAX to calculate the average trading volume over a 20-day period based on the current date context. The DATESINPERIOD function was used to define the required period.

```
Volume 20 SMA =  
VAR CurrentDate = MAX(Dim_Date[Date])  
VAR PeriodDays = 20  
VAR DateInPeriod =  
    DATESINPERIOD(  
        Dim_Date[Date],  
        CurrentDate,  
        -PeriodDays,  
        DAY  
    )  
RETURN  
    CALCULATE(  
        AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]),  
        DateInPeriod  
    )
```

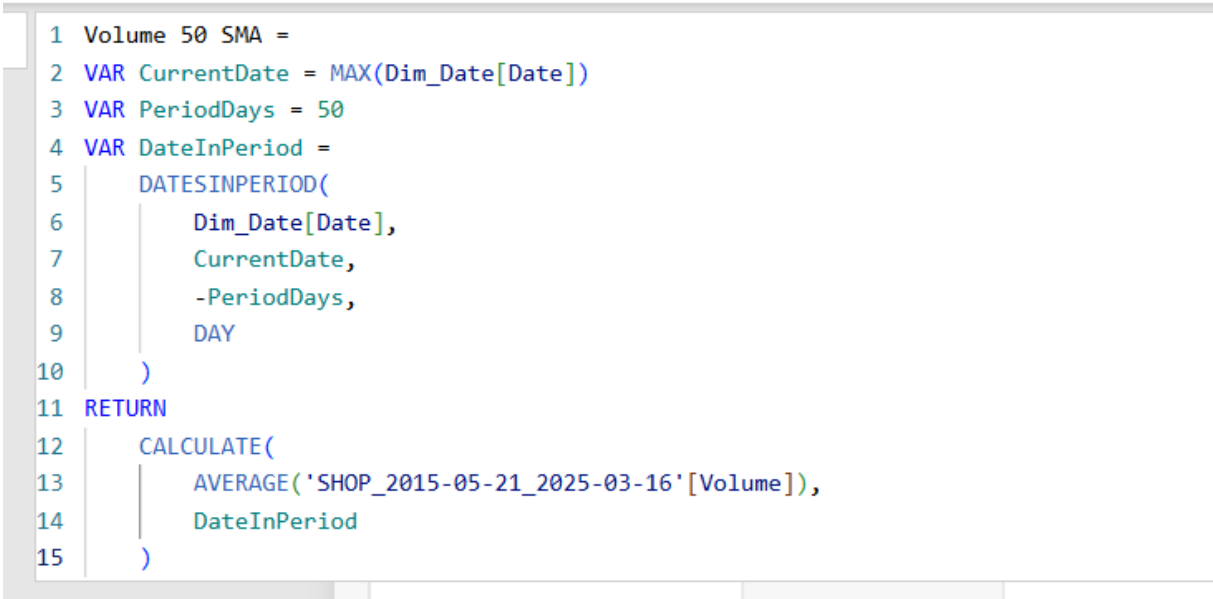


Figure 1: 20-Day Simple Moving Average DAX Measure

3. Calculating 50-Day Simple Moving Average:

A 50-day Simple Moving Average was created using a similar DAX structure. The period was changed to 50 days, allowing the analysis to compare a shorter moving average with a longer moving average.

```
Volume 50 SMA =  
VAR CurrentDate = MAX(Dim_Date[Date])  
VAR PeriodDays = 50  
VAR DateInPeriod =  
    DATESINPERIOD(  
        Dim_Date[Date],  
        CurrentDate,  
        -PeriodDays,  
        DAY  
    )  
RETURN  
    CALCULATE(  
        AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]),  
        DateInPeriod  
    )
```



```
1 Volume 50 SMA =  
2 VAR CurrentDate = MAX(Dim_Date[Date])  
3 VAR PeriodDays = 50  
4 VAR DateInPeriod =  
5     DATESINPERIOD(  
6         Dim_Date[Date],  
7         CurrentDate,  
8         -PeriodDays,  
9         DAY  
10    )  
11 RETURN  
12     CALCULATE(  
13         AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]),  
14         DateInPeriod  
15    )
```

Figure 2: 50-Day Simple Moving Average DAX Measure

4. Moving Average Crossover Signal:

A Moving Average Crossover Signal was created to compare the 20-day SMA with the 50-day SMA. When the 20-day SMA is greater than the 50-day SMA, the signal is classified as Bullish. When the 20-day SMA is lower than the 50-day SMA, the signal is classified as Bearish. Blank values are returned when either moving average is unavailable.

```
MA Crossover Signal =  
IF(  
    ISBLANK([Volume 20 SMA]) || ISBLANK([Volume 50 SMA]),  
    BLANK(),  
    IF(  
        [Volume 20 SMA] > [Volume 50 SMA],  
        "Bullish",  
        IF(  
            [Volume 20 SMA] < [Volume 50 SMA],  
            "Bearish",  
            BLANK()  
        )  
    )
```

```

[Volume 20 SMA] > [Volume 50 SMA],
"Bullish (20 > 50)",
"Bearish (20 < 50)"
)
)

```

The screenshot shows a code editor with a 'structure' pane on the left and a 'formatting' pane on the right. The main editor area contains the following DAX code:

```

1 MA Crossover Signal =
2 IF(
3     ISBLANK([Volume 20 SMA]) || ISBLANK([Volume 50 SMA]),
4     BLANK(),
5     IF(
6         [Volume 20 SMA] > [Volume 50 SMA],
7         "Bullish (20 > 50)",
8         "Bearish (20 < 50)"
9     )
10 )

```

Figure 3: Moving Average Crossover Signal DAX Measure

5. Creating the Price Trend Visualization:

An interactive line and stacked column visual was configured to examine the stock trend over time. The Date field from the Dim_Date table was used on the X-axis with continuous scaling. The visual contains Close Price, Volume 20 SMA and Volume 50 SMA, allowing the movement of the stock price and volume-based moving averages to be viewed together. A zoom slider was also enabled to support closer inspection of different time periods.

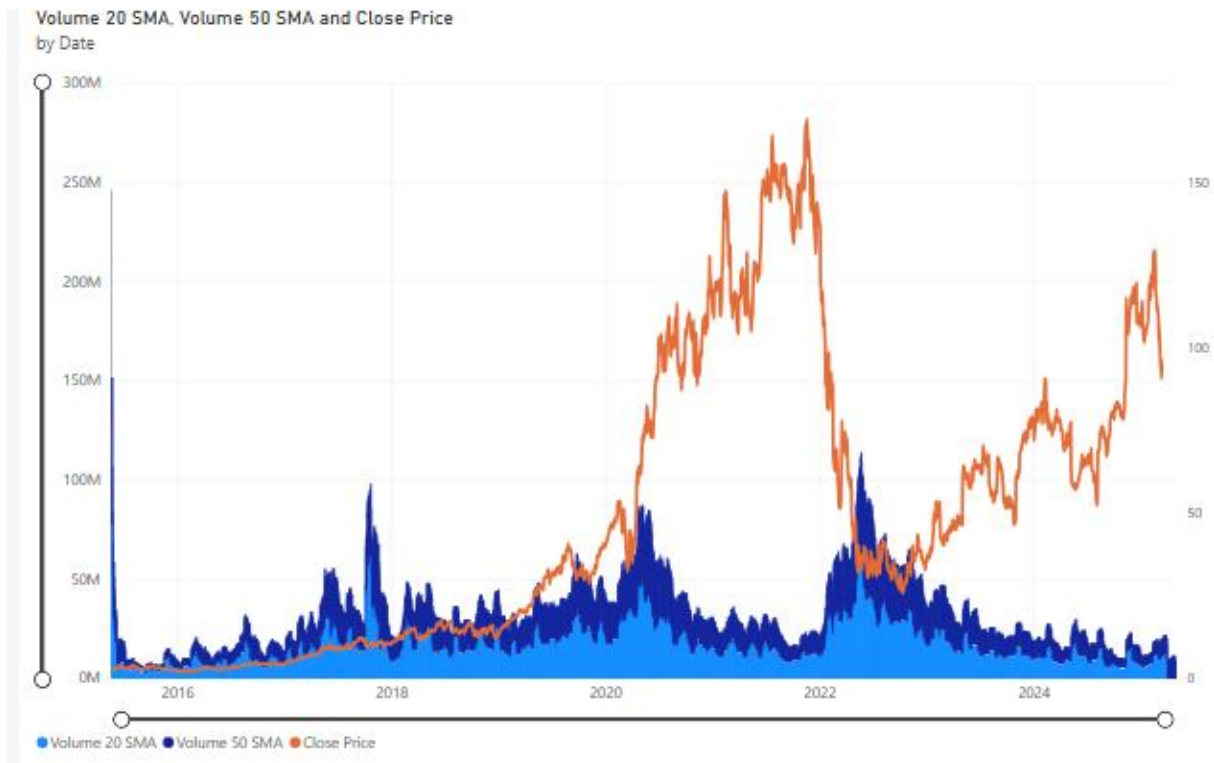


Figure 4: Price Trend and Moving Average Visualization

6. Trading Volume Analysis:

A separate trading-volume visualization was created to examine changes in trading activity over time. The visual uses the date dimension along with Trading Volume and the 20-day SMA. This helps identify periods where trading activity increases and compare the observed volume with its moving-average trend.

The completed dashboard also provides an interactive view of the trading-volume series alongside the main price trend, making it possible to inspect changes in trading activity across the available period.

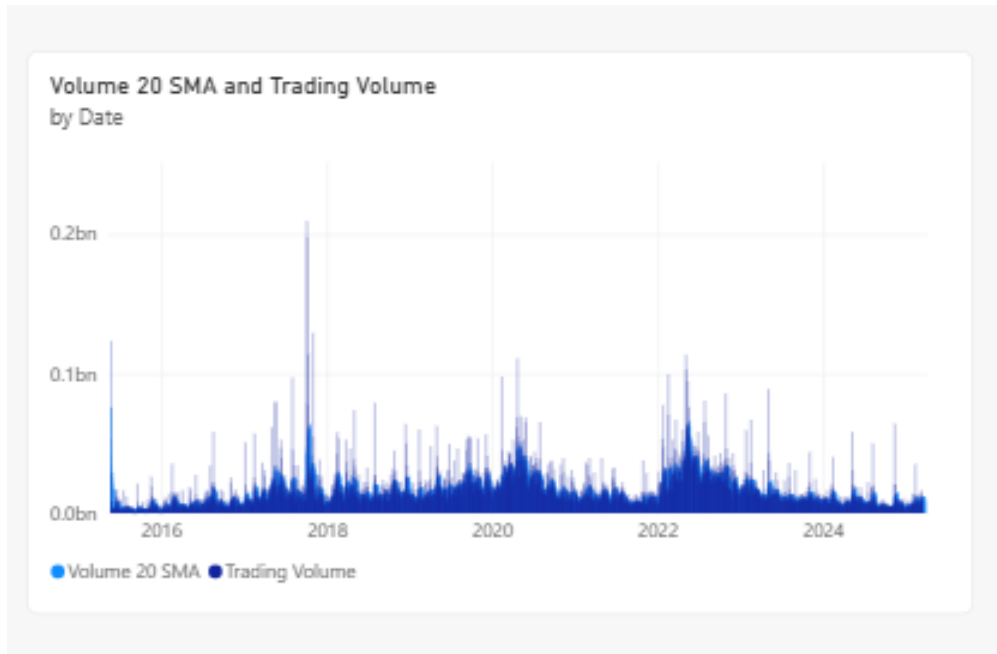


Figure 5: Trading Volume and 20-Day SMA Visualization

7. Configuring Timeline Slicers and Dynamic Parameter:

a) Date Range Slicer

A date-range slicer was added to allow the user to select a specific period for analysis. The completed dashboard shows a date range from 01-01-2015 to 31-12-2025, with slider controls for changing the selected interval.



Figure 6: Date Range Slicer

b) Dynamic Parameter Selector

A field parameter was configured to allow different price metrics to be selected dynamically. The available parameter fields include Close Price, High Price, Low Price and Open Price. This provides a flexible way to change the metric used for analysis without creating separate visuals for every price field.

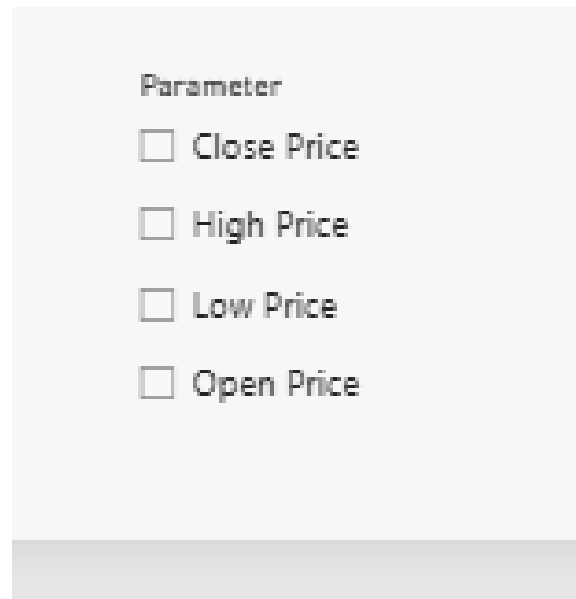


Figure 7: Dynamic Parameter Selector

8. Creating the Custom Tooltip:

A custom tooltip page named `Stock_Tooltip` was created to provide detailed information when a user hovers over a selected point in the main visualization. The tooltip displays Open Price, High Price, Low Price, Close Price, Daily Return % and Trading Volume.

The tooltip provides instantaneous stock information in a compact format and allows the main dashboard to remain visually simple while still giving access to detailed values.

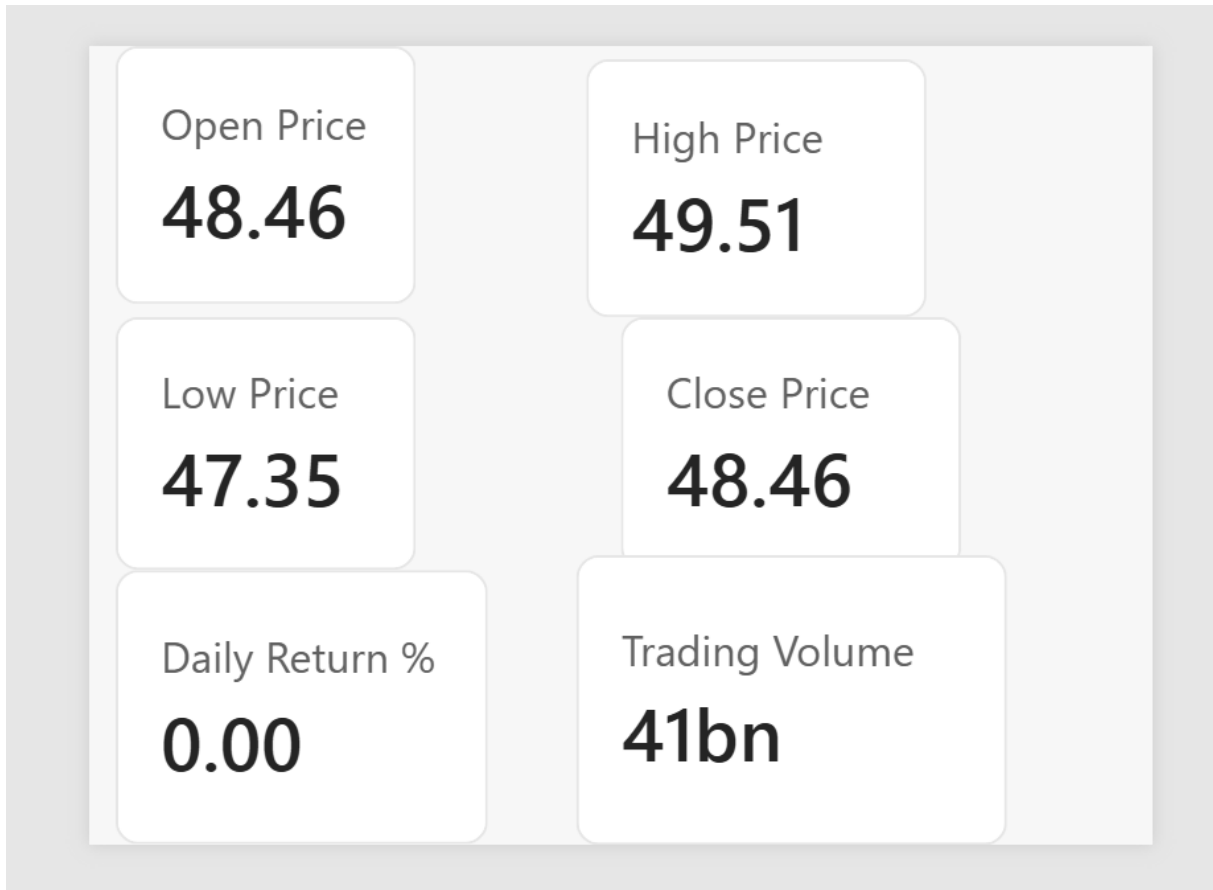


Figure 8: Custom Stock Tooltip Displaying Detailed Metrics

9. Final Power BI Dashboard:

The completed Power BI dashboard combines the moving-average calculations, stock price visualization, trading-volume analysis, date filtering and dynamic parameter selection into a single interactive report page. The dashboard contains the main price and moving-average visual, a trading-volume visual, date controls and parameter controls.

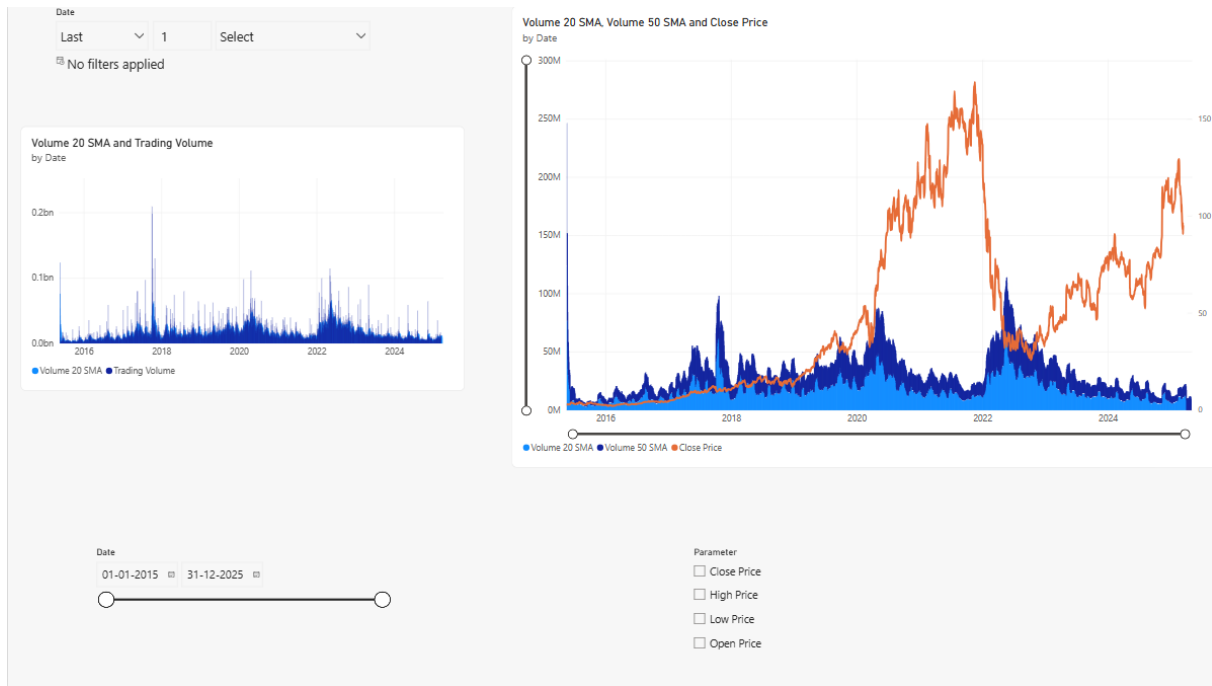


Figure 9: Final Interactive Shopify Stock Analysis Dashboard

10. Conclusion:

Week 4 provided practical experience in performing Exploratory Data Analysis on Shopify stock data using Microsoft Power BI. The 20-day and 50-day Simple Moving Averages were calculated using DAX, and a moving average crossover signal was developed to classify the relationship between the two moving averages as bullish or bearish.

Interactive visualizations were created to examine the stock price trend and trading volume over time. Timeline filtering and a dynamic parameter selector were configured to support flexible analysis. A custom tooltip was also implemented to display Open Price, High Price, Low Price, Close Price, Daily Return % and Trading Volume for detailed inspection of selected data points.

Overall, the completed dashboard provides an interactive environment for exploring Shopify stock price and trading-volume data and demonstrates the use of DAX calculations, Power BI visualizations and interactive reporting features.